

Physics 201 - Lecture 17

→ More Assignment 5 😊

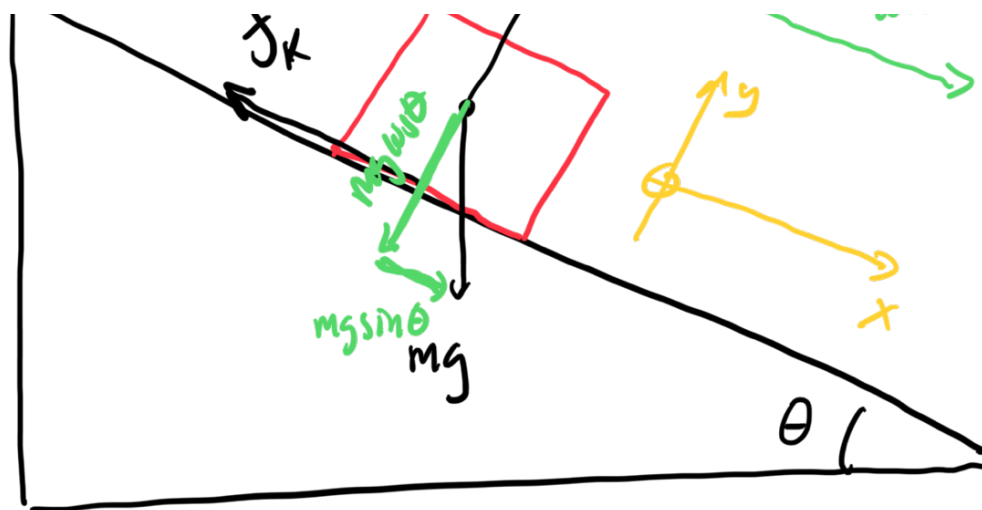
→ These questions are going to get more challenging as the assignment progresses, but.....

IF WE FOLLOW THE PROCESS,
WE CAN SOLVE THEM IN
THE SAME WAY AS FOR
THE EASIER PROBLEMS

3.

↑ N

Motion



① FBD

- (i) identify forces
- (ii) choose coordinate system
- (iii) resolve into components

friction { static / kinetic ?
direction ?

②

$$\Sigma F_x = ma_x$$

$$mg \sin \theta - f_k = ma_x$$

$$\Sigma F_y = ma_y = 0$$

$$N - mg \cos \theta = 0$$

$$N = mg \cos \theta$$

$$f_k = \mu_k N$$

$$\cancel{mg \sin \theta} - \mu_k \cancel{mg \cos \theta} = \cancel{max} \quad | \quad f_k = \mu_k mg \cos \theta$$

$$a_x = g \left(\sin \theta - \mu_k \cos \theta \right)$$

$$= 1.5 \text{ m/s}^2$$

b) "constant velocity" $\rightarrow \vec{a}^0 = 0$

$$\boxed{a_x = 0}, \quad a_y = 0$$

$$0 = g \left(\sin \theta - \mu_k \cos \theta \right)$$

$$= 0$$

$$\sin \theta = \mu_k \cos \theta$$

$$\sin \theta = \mu_K \cos \theta$$

$$\boxed{\mu_K = \tan \theta}$$

0.100

$$\mu_K \cos \theta = \sin \theta$$

$$\mu_K = \frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$\theta = \tan^{-1} \left(\mu_K \right)$$

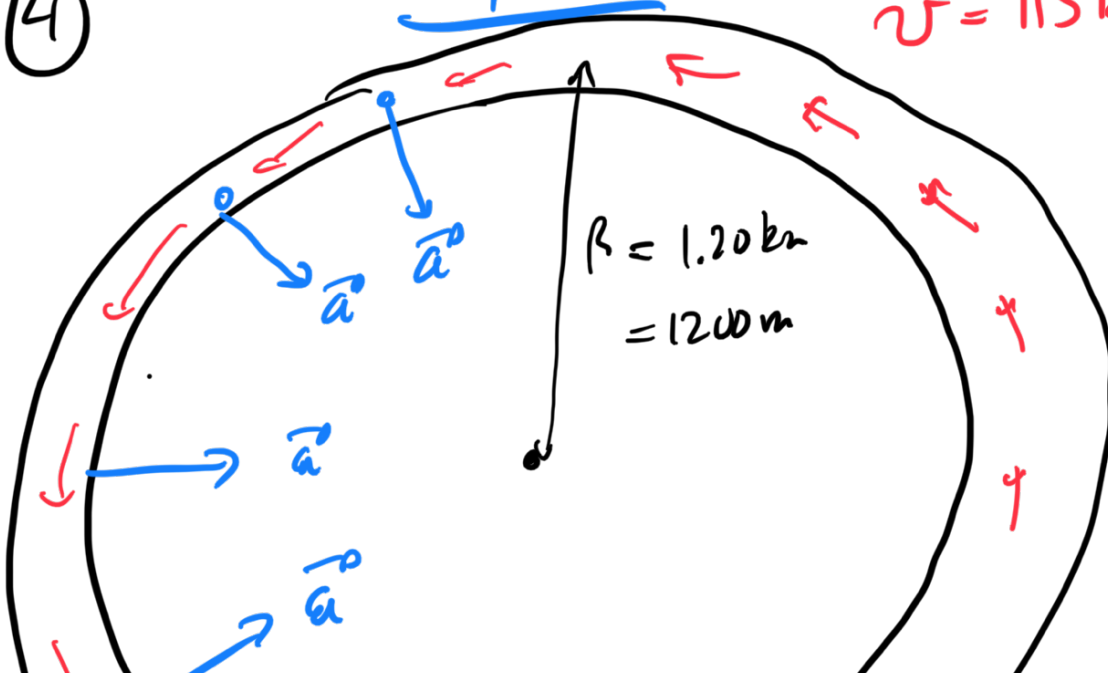
0.100

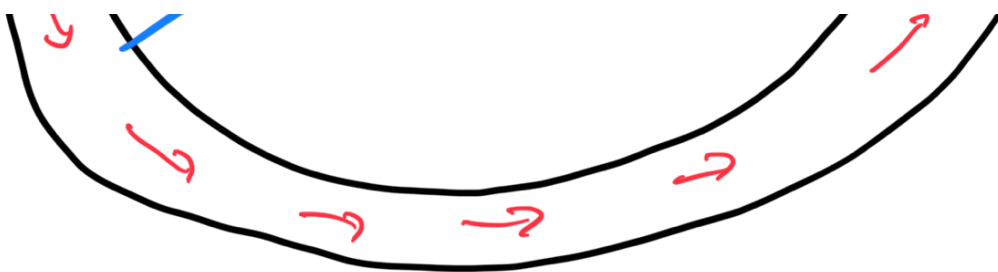
$$\boxed{\theta = 5.71^\circ}$$

(4)

Top View

$$v = 115 \text{ km/h}$$





"Uniform circular motion"

$$|\vec{a}| = \frac{v^2}{R} = \frac{(31.94)^2}{1200} = 0.850 \text{ m/s}^2$$

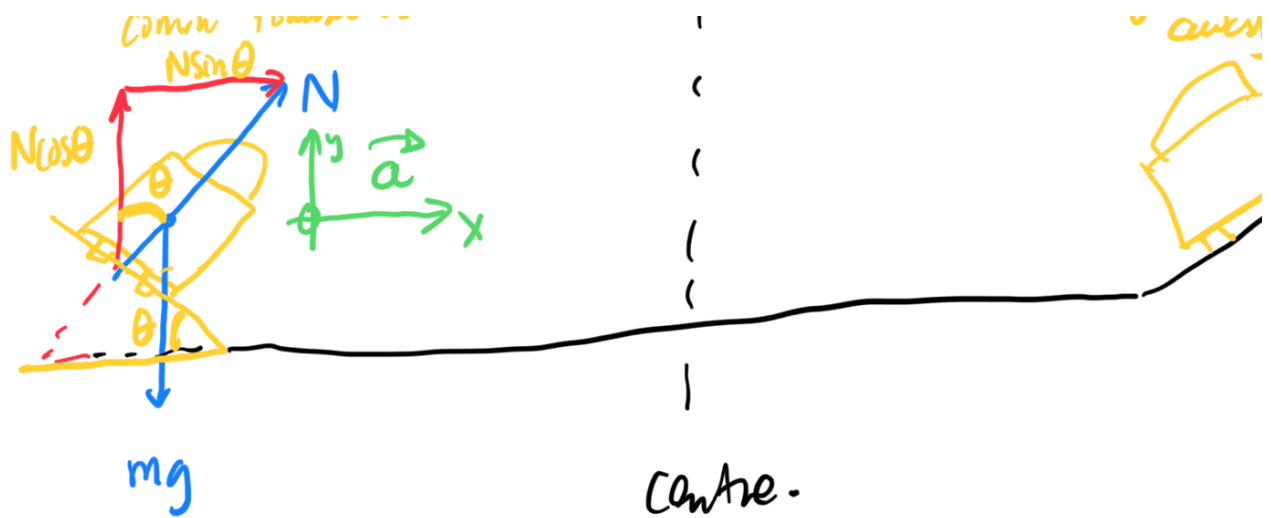
$$v = 115 \frac{\text{km}}{\text{hr}} \times \frac{1000 \text{ m}}{\text{km}} \times \frac{1 \text{ hr}}{3600 \text{ s}} = 31.94 \text{ m/s} \leftarrow$$

Side View

"ideal banking angle"
↳ no friction!

... hands vs

going



FBD - identify forces
 → coordinate system.

such that x -axis is
 in the direction of the
 acceleration.

→ resolve into components.

x	y
$a_x = 0.850 \text{ m/s}^2$	$a_y = 0$
$N \sin \theta = m a_x$	$N \cos \theta - mg = 0$

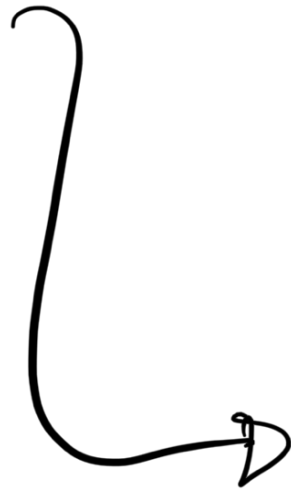
mg


$$N = \frac{mg}{\cos \theta}$$

$$\left(\frac{mg}{\cos \theta} \right) \sin \theta = ma_x$$

$$g \tan \theta = a_x$$

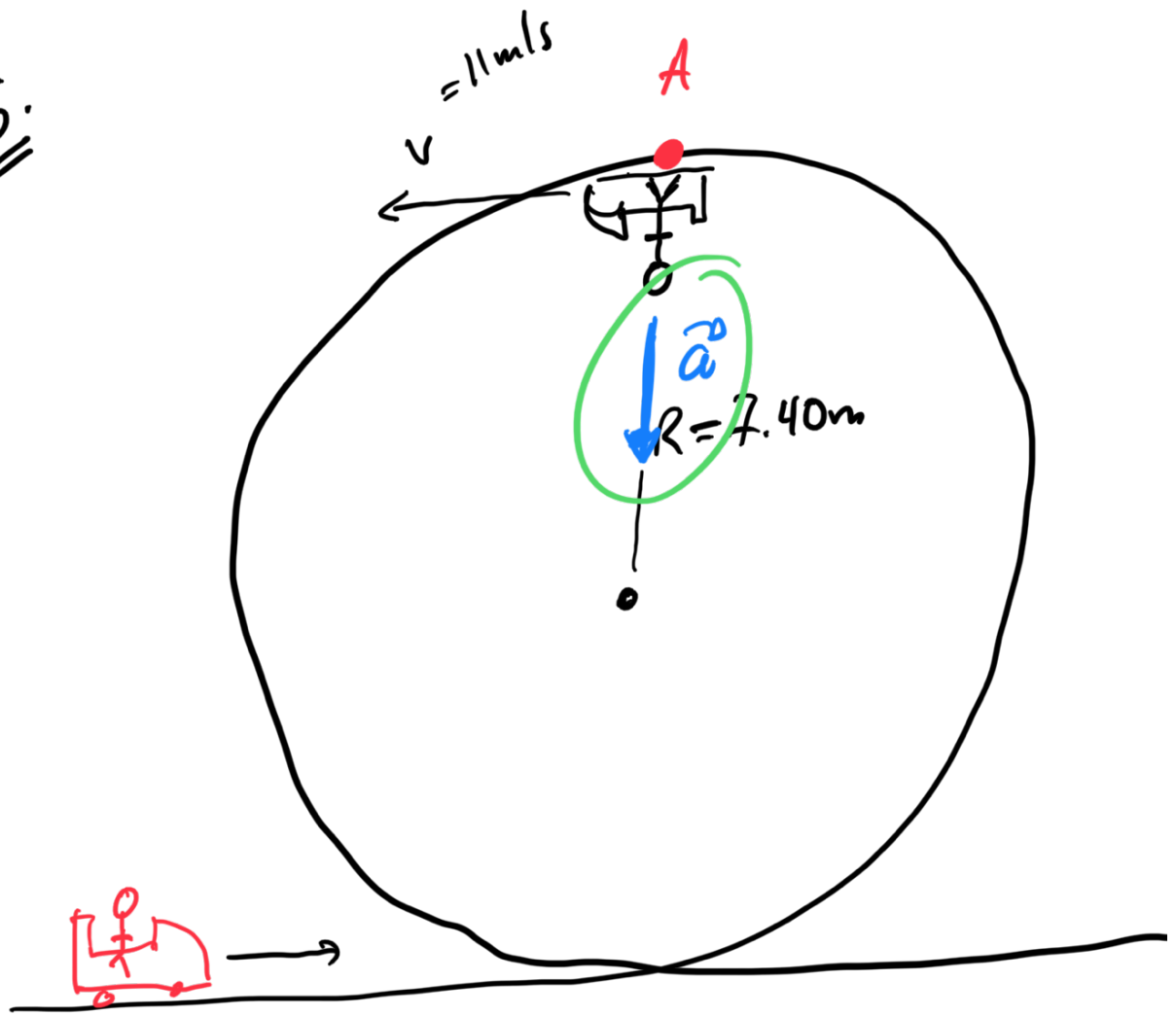
$$\tan \theta = \frac{a_x}{g}$$



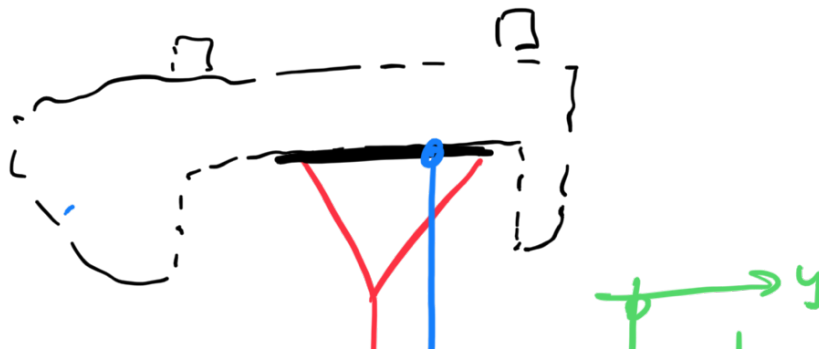
$$\theta = \tan^{-1} \left(\frac{a_x}{g} \right)$$

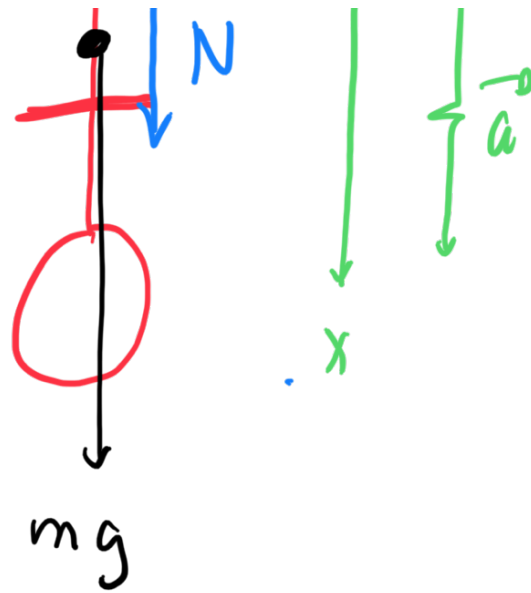
$$= 4.96^\circ$$

5.



$$|\vec{a}|_c = \frac{v^2}{R} = \frac{(11.0)^2}{7.4} = 16.35 \text{ m/s}^2$$





$$mg + N = ma$$

$$N = ma - mg$$

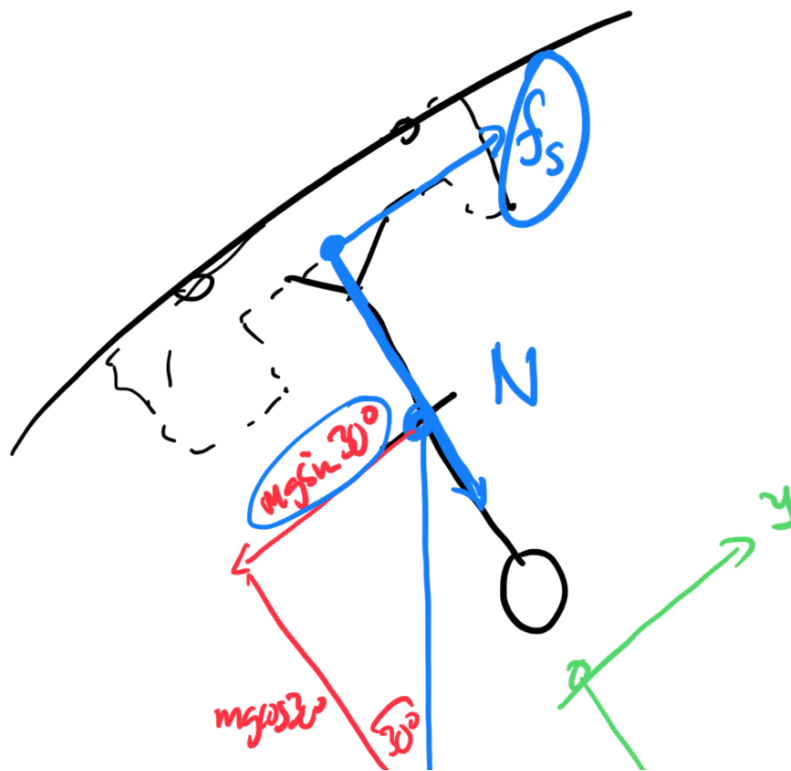
$$= m(a - g)$$

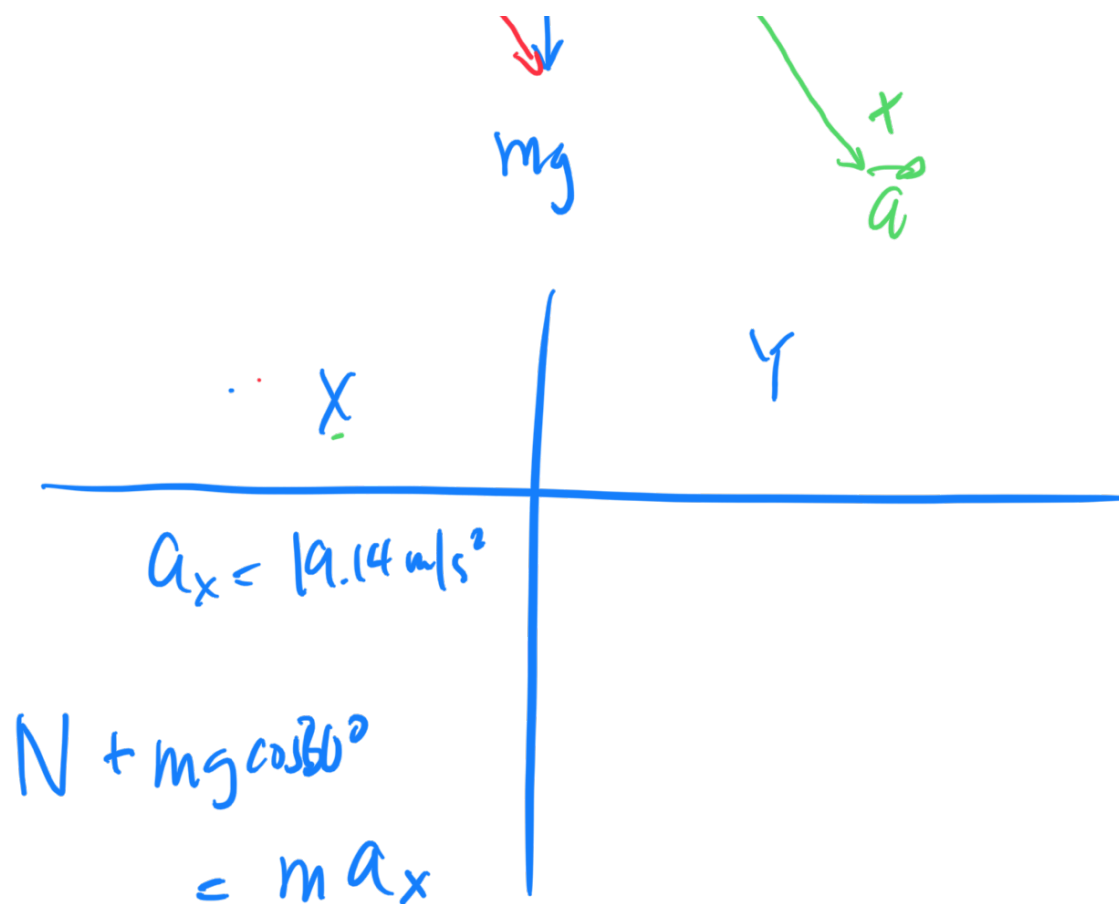
$$= 40 \text{ kg} (16.35 - 9.8)$$

$$|\vec{N}_{\text{normal}}| = 262 \text{ N}$$



$$|\vec{a}_c| = \frac{v^2}{R} = 19.14 \text{ m/s}^2$$





Free-body diagram showing forces and acceleration:

- Force mg (gravity) acting vertically downwards.
- Acceleration a acting diagonally downwards and to the right.
- Coordinate system with x (horizontal) and y (vertical) axes.

Equation for the x-direction:

$$a_x = 19.14 \text{ m/s}^2$$

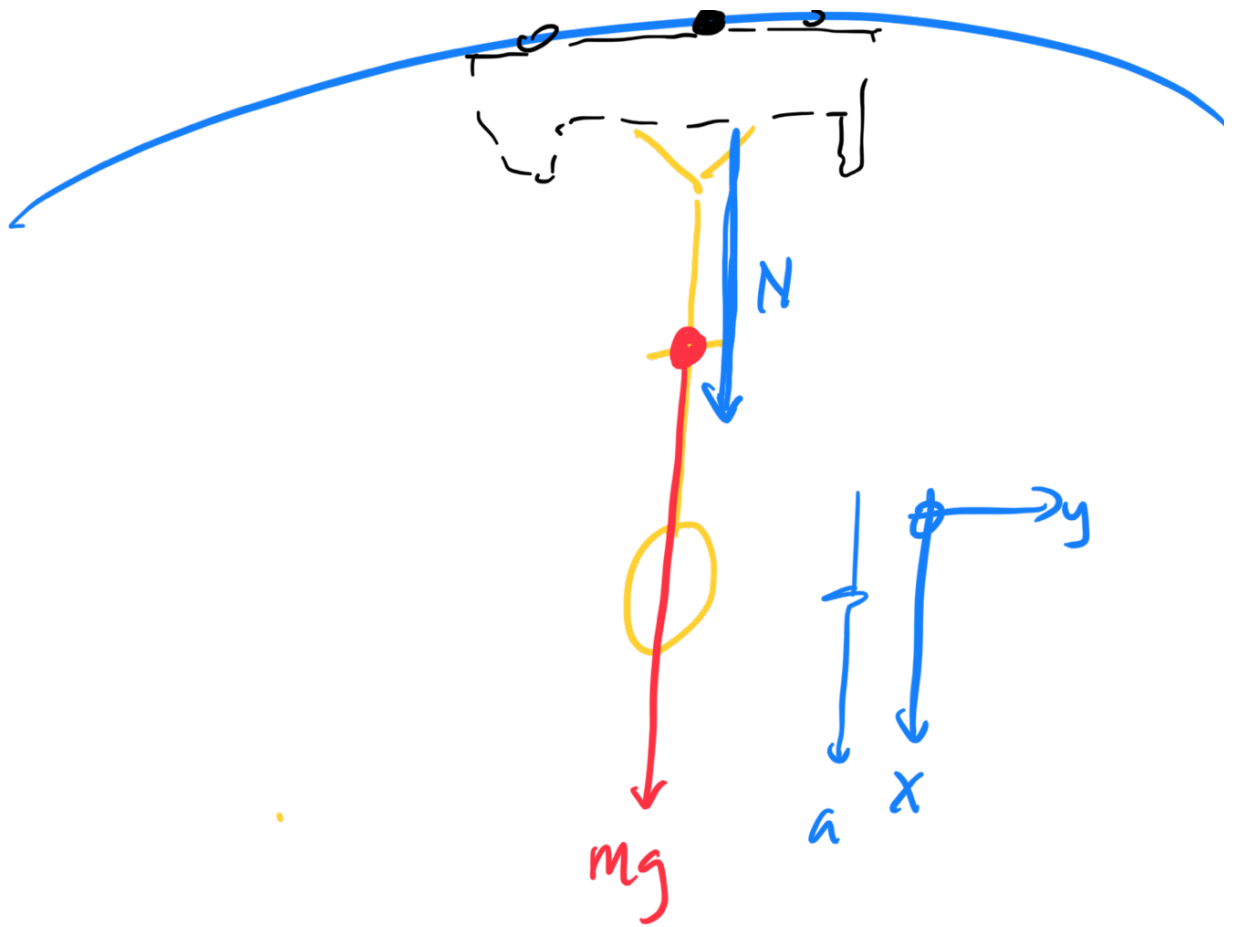
$$N + mg \cos 30^\circ = ma_x$$

$$N = ma_x - mg \cos 30^\circ$$

$$= (40)(19.14) - (40)(9.8)(\cos 30^\circ)$$

$$N = 426 \text{ N}$$

A



$$\cancel{N + mg} = \cancel{max}$$

When do you fall out of your seat?

$$N = 0 !$$

$$(a_x) = g \quad \text{falling !}$$



$$\frac{v^2}{R} = g$$

$$v^2 = g R$$

$$v = \sqrt{g R}$$

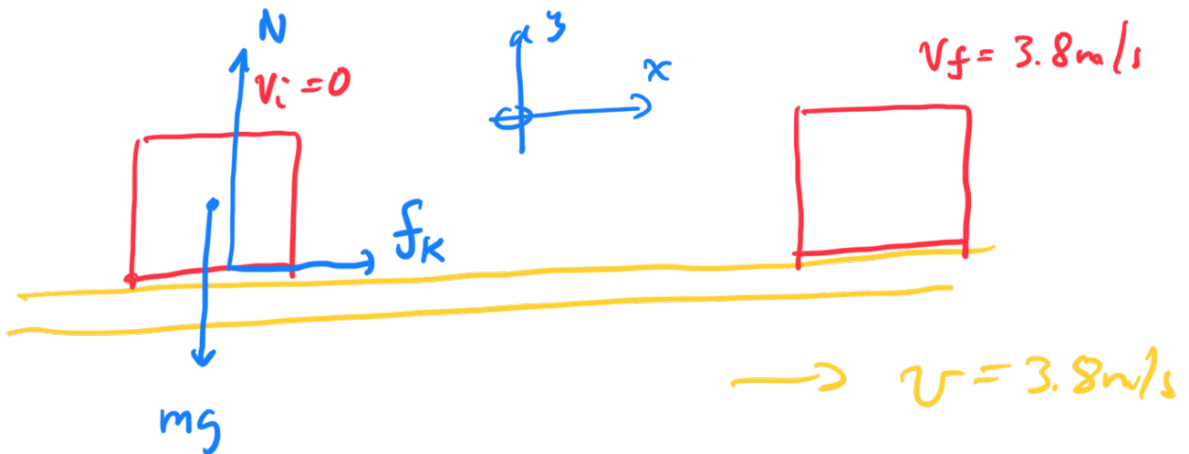
$$= \sqrt{(9.8)(7.4)}$$

$$v = 8.52 \text{ m/s}$$

close !!!

"weightless news"

6.



$$f_k = m a_x$$

$$N - mg = 0$$

$$N = mg$$

$$\mu_k mg = m a_x$$

$$f_k = \mu_k N = \mu_k mg$$

$$\boxed{a_x = \mu_k g} = (0.25)(9.8) = 2.45 \text{ m/s}^2$$

$$v_i = 0$$

$$v_f = 3.8 \text{ m/s}$$

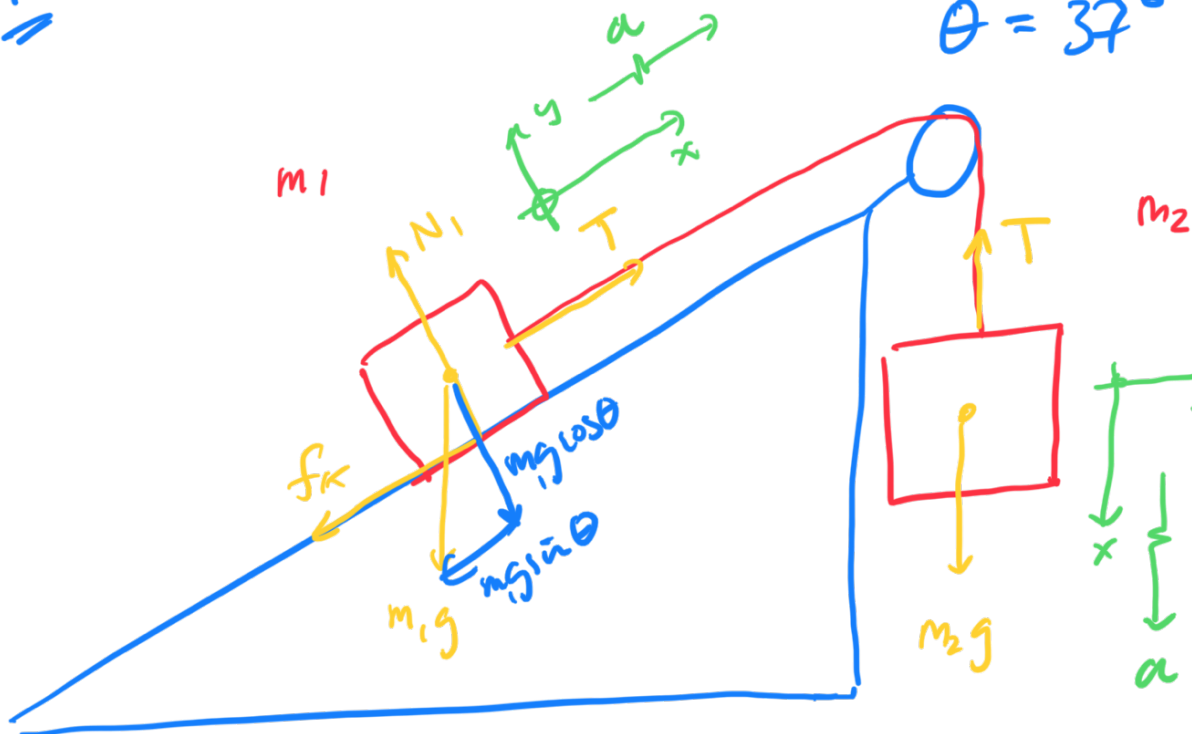
$$a = 2.45 \text{ m/s}^2$$

$$t = ?$$

$$v_f = v_i + at$$

$$t = \frac{v_f}{a} = \frac{3.8 \text{ m/s}}{2.45 \text{ m/s}^2} = \underline{\underline{1.55 \text{ s}}}$$

7. $m_1 = 3.0 \text{ kg}$, $m_2 = 7.0 \text{ kg}$, $\mu_k = 0.46$
 $\theta = 37^\circ$



$m_2 > m_1$, therefore system must move

to the right.

X	Y
$1:$ $T - m_1 g - f_K = m_1 a$	$N_1 - m_1 g \cos \theta = 0$ $N_1 = m_1 g \cos \theta$

$f_K = \mu_K m_1 g \cos \theta$

$$T - m_1 g - \mu_K m_1 g \cos \theta = m_1 a$$

$$2: \quad m_2 g - T = m_2 a$$

ADD!

$$m_2 g - m_1 g - \mu_K m_1 g \cos \theta = (m_1 + m_2) a$$

$$a = g \left[\frac{m_2 - m_1 - \mu_k m_1 \cos \theta}{m_1 + m_2} \right]$$

$$a = 4.01 \text{ m/s}^2$$